July 2025 CSE 208
Online on APSP

Time: 30 minutes

Arbitrage is the use of discrepancies in currency exchange rates to transform one unit of a currency into more than one unit of the same currency. A currency is said to **admit arbitrage** if it is possible to start with one unit of that currency, perform a sequence of currency exchanges, and end with more than one unit of the original currency.

For example, suppose that:

- 1 US Dollar buys 1.5 Australian Dollars,
- 1 Australian Dollar buys 81.85 Bangladeshi Taka, and
- 1 Bangladeshi Taka buys 0.0082 US Dollars.

Then, by converting currencies, a trader can start with 1 US Dollar and obtain
(1.5 * 81.85 * 0.0082 = 1.007) US Dollars, making a profit.
Thus, the currency **USD** admits arbitrage.

Your task is to write a program that determines **all currencies** for which arbitrage is possible.

Input

The first line contains an integer n ($1 \le n \le 100$), representing the number of different currencies.

The next n lines each contain the name of one currency.
Each currency name consists of a single string with no spaces.

The next line contains an integer m ($1 \le m \le n \times n$), representing the number of available exchange rates.

The following m lines each contain:

- the name c_i of a source currency,
- a real number r_{ij} representing the exchange rate from c_i to c_j,
- and the name c_j of the destination currency.

Note that c_i and c_j may be the same currency.
Any exchange not listed in the input is considered impossible.

Output

Print the names of **all currencies** for which arbitrage is possible, one per line, in the order in which they appear in the input.

If no currency admits arbitrage, print a single line containing:

```
No Arbitrage
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Example

Sample Input	**Sample Output**
3 USD AUD BDT 3 USD 1.5 AUD AUD 81.85 BDT BDT 0.0082 USD	USD AUD BDT
3 USD AUD BDT 6 USD 1.5 AUD USD 122.36 BDT AUD 81.85 BDT AUD 0.66 USD BDT 0.012 AUD BDT 0.0081 USD	No Arbitrage
3 EUR USD GBP 4 EUR 0.5 USD USD 0.5 EUR USD 2.0 GBP GBP 1.1 USD	EUR USD GBP

Explanation of Sample 3

USD: USD → GBP → USD (1.0×2.0×1.1=2.2)
GBP: GBP → USD → GBP (1.0×1.1×2.0=2.2)
EUR: EUR → USD → GBP → USD → GBP → USD → EUR (1.0×0.5×2.0×1.1×2.0×1.1×0.5=1.21)